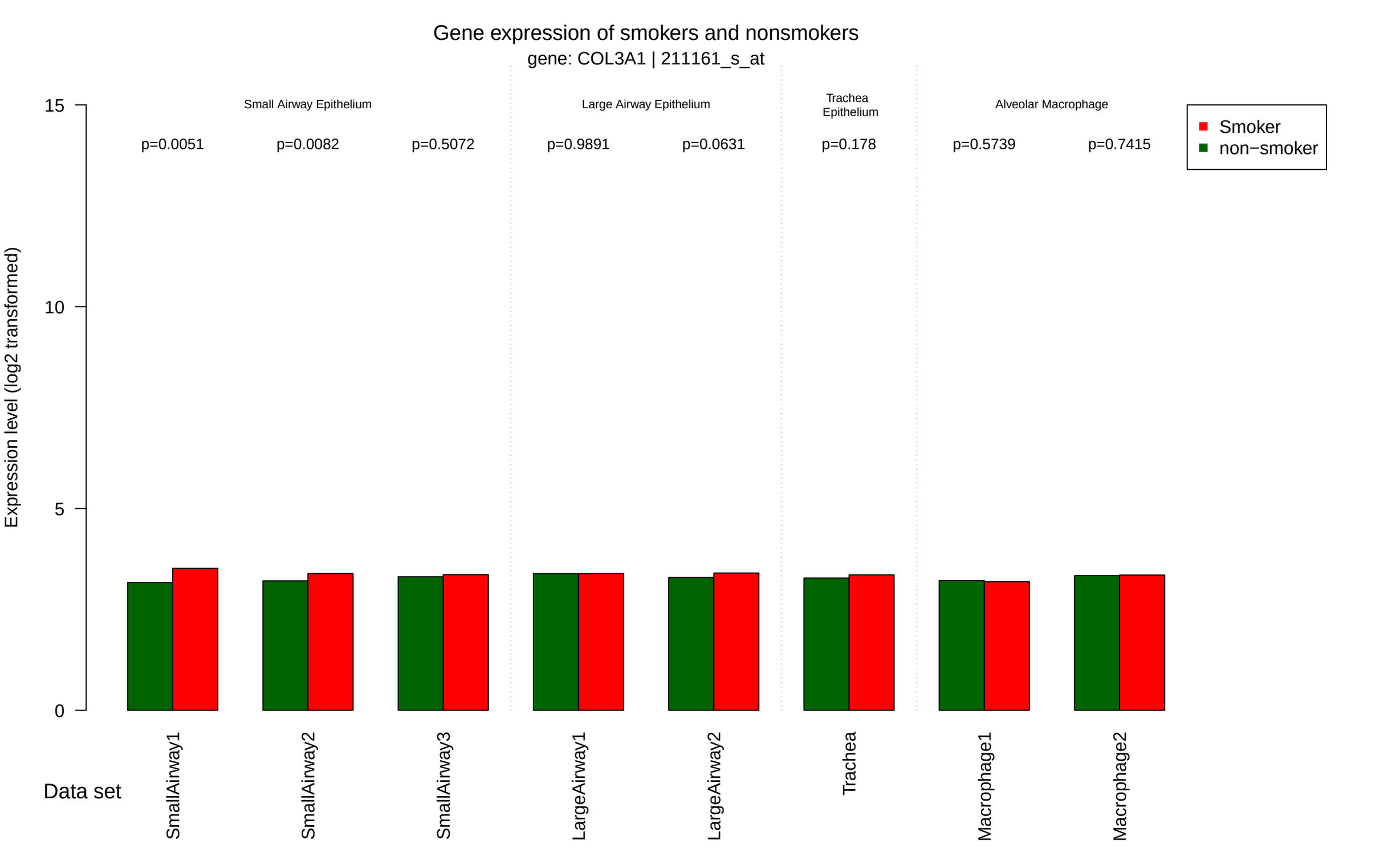
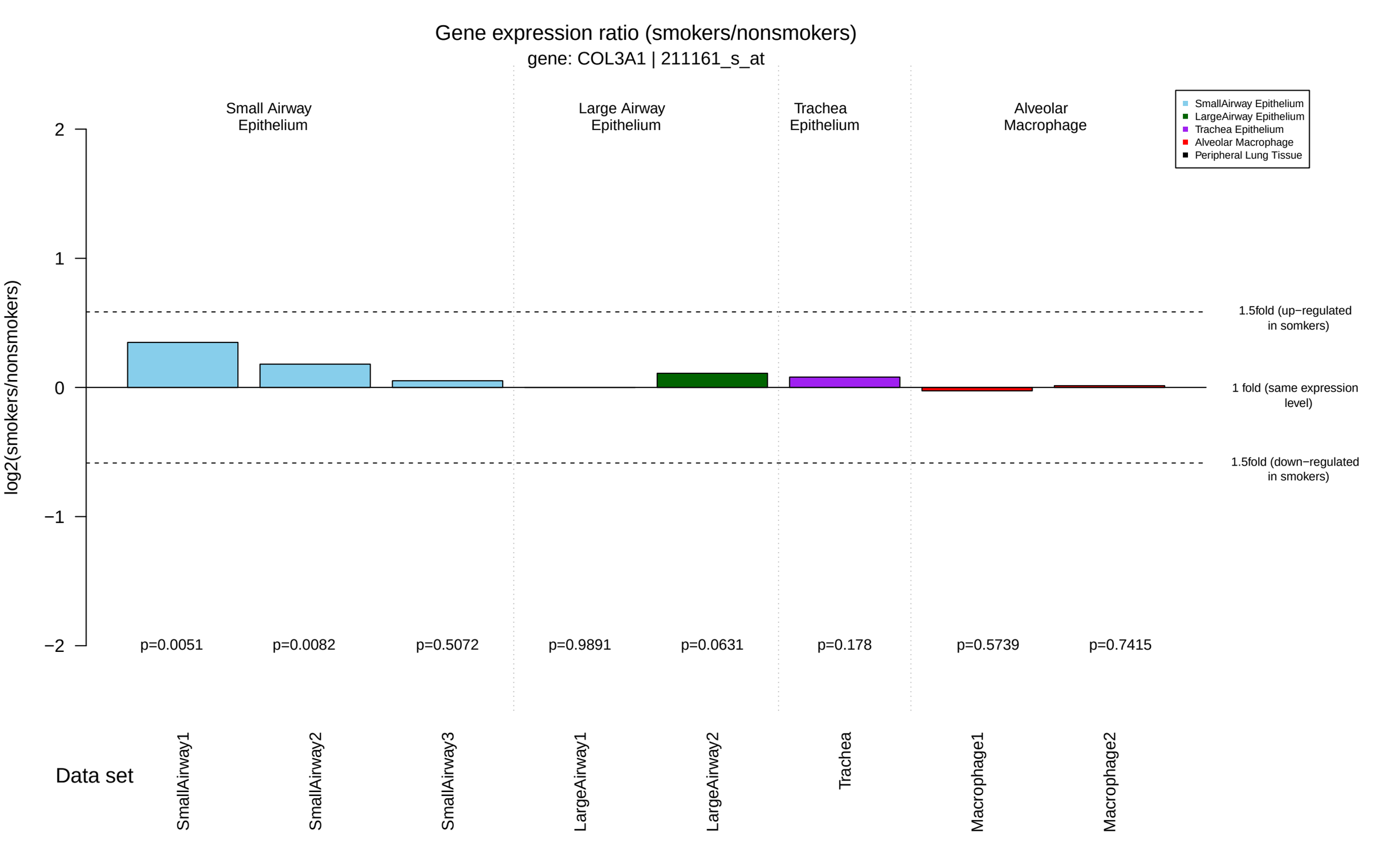
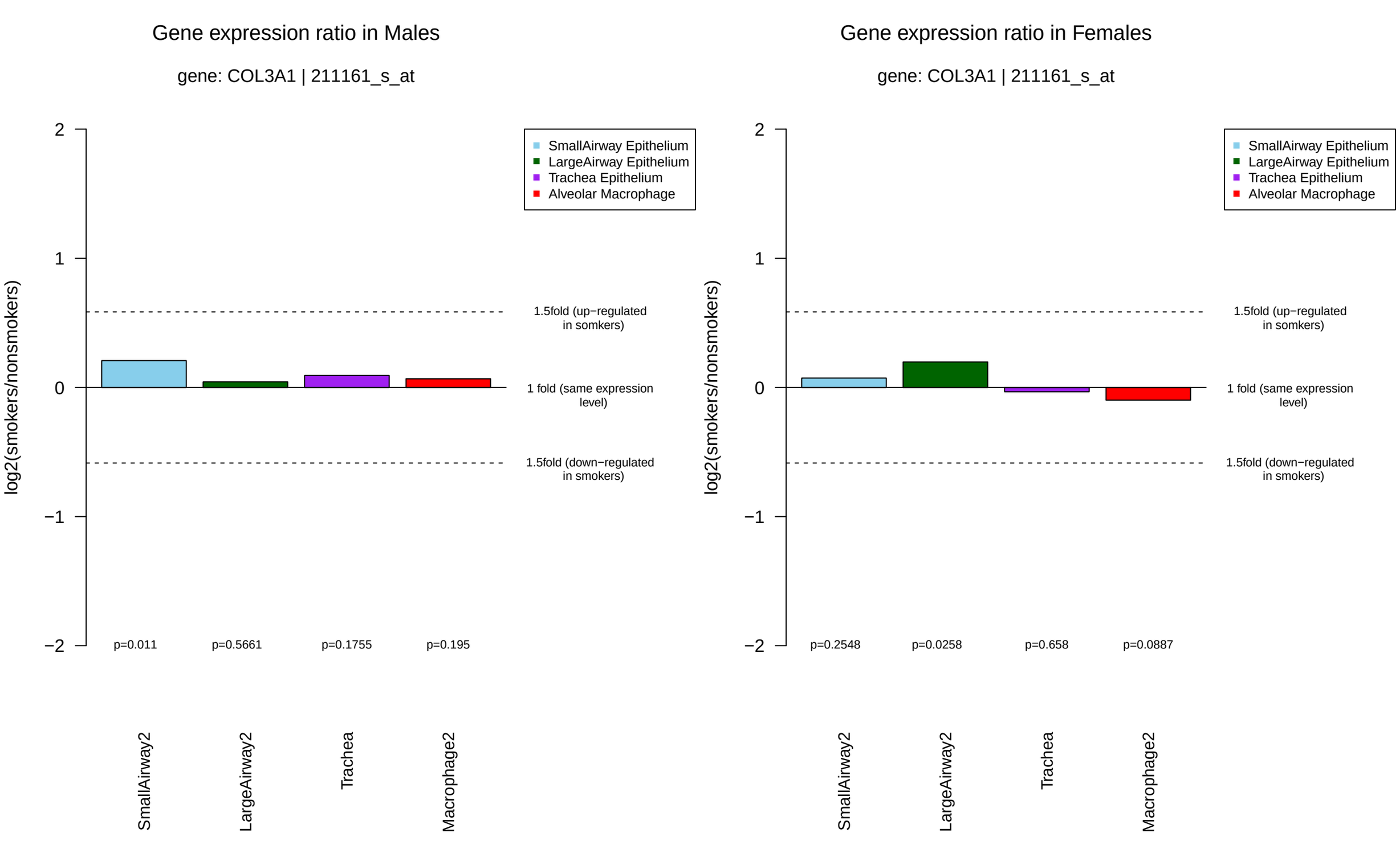
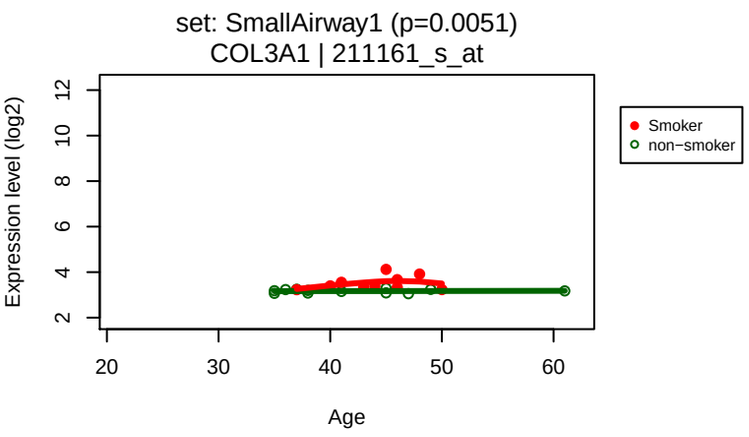
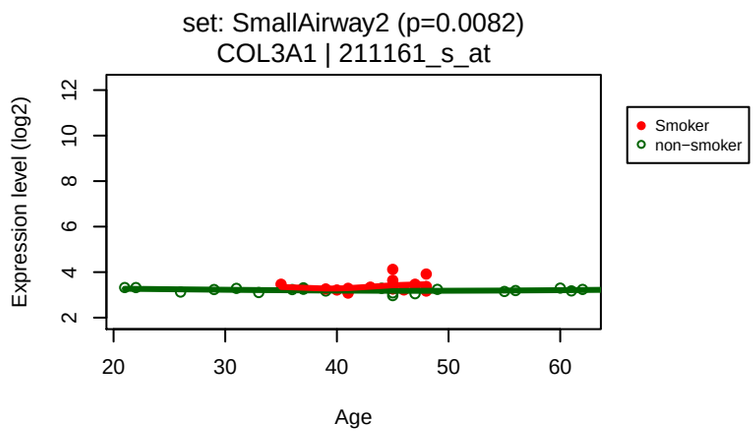
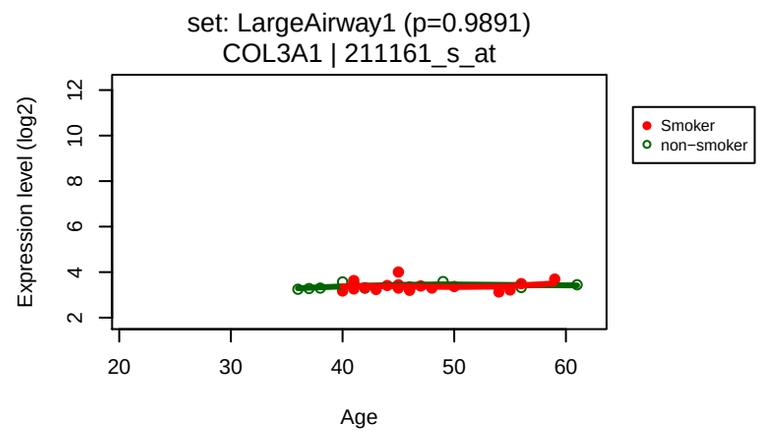

Gene expression of smokers and nonsmokers
gene: COL3A1 | 211161_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0051
p=0.0082
p=0.5072
p=0.9891
p=0.0631
p=0.178
p=0.5739
p=0.7415
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

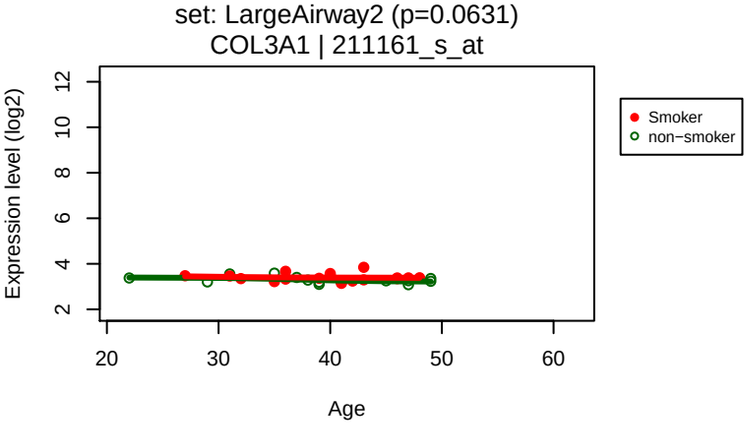

Gene expression ratio (smokers/nonsmokers)
gene: COL3A1 | 211161_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0051
p=0.0082
p=0.5072
p=0.9891
p=0.0631
p=0.178
p=0.5739
p=0.7415
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

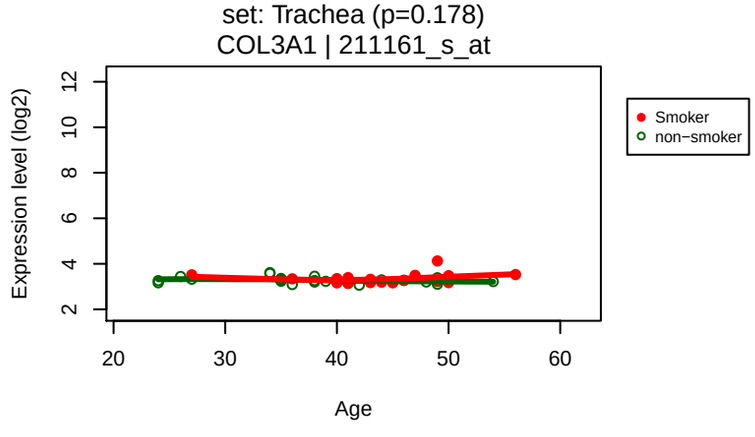

Gene expression ratio in Males
gene: COL3A1 | 211161_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.011
p=0.5661
p=0.1755
p=0.195
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL3A1 | 211161_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2548
p=0.0258
p=0.658
p=0.0887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

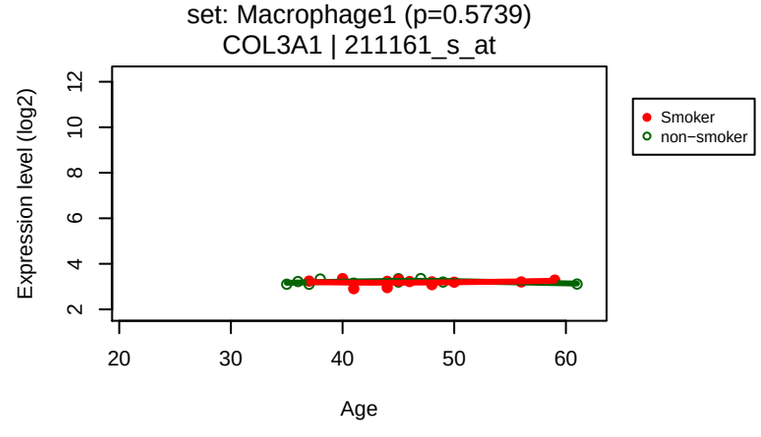

set: SmallAirway1 (p=0.0051)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker

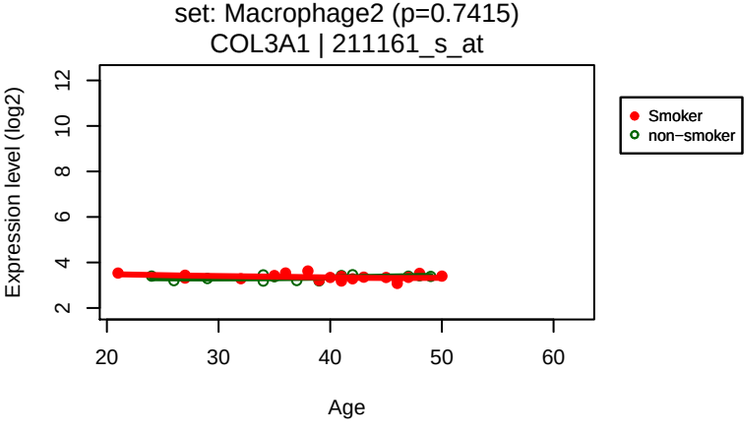

set: SmallAirway2 (p=0.0082)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker

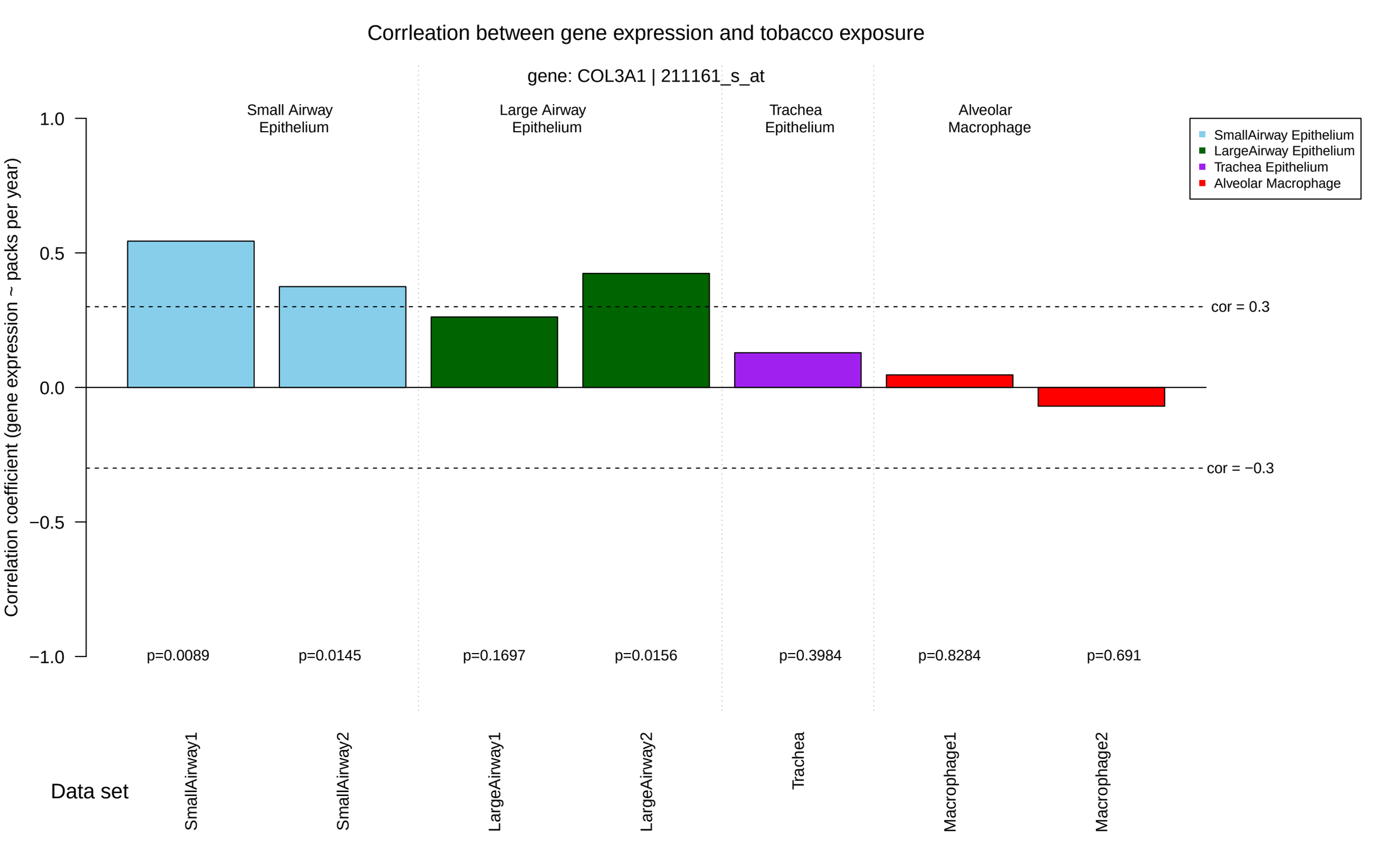

set: LargeAirway1 (p=0.9891)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0631)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.178)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5739)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7415)
COL3A1 | 211161_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL3A1 | 211161_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0089
p=0.0145
p=0.1697
p=0.0156
p=0.3984
p=0.8284
p=0.691
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2